NAZIR NAYAL

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EDUCATION

Universität des Saarlandes Sept 2024 - Aug 2027
Doctor of Philosophy - PhD, Computer Science. *Saarbrücken, Saarland, Germany*
- Part of the CS@Max Planck Doctoral Program

Koç University Sept 2021 - Sept 2023
Master of Science, Computer Science and Engineering. *Istanbul, Turkey*
- Funded by KUIS AI Center and Newton Advanced Fellowship (Royal Society UK)

Koç University Sept 2017 - June 2021
Bachelor of Science, Computer Engineering. *Istanbul, Turkey*
- Full scholarship funded by Al Ghurair Foundation (AGFE).

RESEARCH INTERESTS

- Generative Modeling, World Modeling, Uncertainty, Multi-Modality.

PUBLICATIONS

- N. Nayal, C. Wewer. J. E. Lenssen "MixFlow: Mixed Source Distributions Improve Rectified Flows" in ICLR 2026 2nd DeLTa Workshop
- **N. Nayal**, Y. Shoeb, F. Güney "A Likelihood Ratio-Based Approach to Segmenting Unknown Objects" in IJCV 2025
- Y. Shoeb, **N. Nayal**, A. Nowzad, F. Güney, H. Gottschalk "Segment-Level Road Obstacle Detection Using Visual Foundation Model Priors and Likelihood Ratios" in VISAPP 2025
- **N. Nayal**, M. Yavuz, J. F. Henriques F. Güney "RbA: Segmenting Unknown Regions Rejected by All" in ICCV 2023.
- Y. Hassanzadeh-Nazarabadi, **N. Nayal**, S. Sameh Hamdan, Ö. Özkasap, and A. Küpçü, "A containerized proof-of-concept implementation of lightchain system," in ICBC. IEEE, 2020.
- Y. Hassanzadeh-Nazarabadi, **N. Nayal**, S. Sameh Hamdan, A. Utkan Şahin, Ö. Özkasap, and A. Küpçü, "Demo: Skip Graph Middleware Implementation", in SRDS 2020.

EXPERIENCE

Max Planck Institute of Informatics (MPI-INF) Sept 2024 - Present
Doctoral Researcher *Saarbrücken, Saarland, Germany*
- Working with the Computer Vision and Machine Learning Group

Visual Geometry Group (VGG), University of Oxford Sept 2023 - Oct 2023
Graduate Short Term Student Visitor *Oxford, United Kingdom*
- Visited VGG as part of our project collaboration with João F. Henriques.

Koç University & İş Bankası Artificial Intelligence Center (KUIS AI) Sept 2021 - Sept 2024
Research Fellow *Istanbul, Turkey*
- Worked on detection of rare events and unknown objects in Autonomous Driving scenarios.

Unvest R&D Center April 2022 - Sept 2024
Research Engineer *Istanbul, Turkey*
- Explored the potential of deploying uncertainty estimation models for autonomous driving.

Syrian Youth Empowerment (SYE) July 2021 - Present
Grad Student Mentor *Remote*
- Mentoring and providing support to Syrian students applying for graduate degrees abroad.

Hospital on Mobile (now CareX.ai) Jan 2021 - June 2021
AI Engineer *Remote*
- Conducted research and developed AI models for vital signs prediction.

Invent Analytics July 2020 - Sept 2020
Software Engineering Intern *Istanbul, Turkey*
- Implemented a machine learning forecasting model based on supermarket products sales data.

Distributed Systems and Reliable Networks Laboratory (DISNET) Mar 2019 - Jan 2021
Research Intern *Istanbul, Turkey*
- Implemented Proof of Concept for LightChain, and an augmented version of Skip Graph Data Structure.

Dapper Labs (Based in Vancouver, Canada) Aug 2019 – Dec 2019
Software Engineering Intern *Remote*
- Contributed to the implementation of the first version of the networking layer of Flow Blockchain.

ACHIEVEMENTS

Al Ghurair Foundation for Education STEM Scholarship (AGFE) Sept 2017 - Sept 2021
Recipient *Istanbul, Turkey*
- A merit-based award for highly achieving Arab students to pursue a degree in STEM.

Vehbi Koç Scholar Award 2018 - 2019 - 2020 - 2021
Award Recipient *Istanbul, Turkey*
- Received this award for 4 consecutive years for maintaining a semester GPA of 3.5+.

IEEEXtreme 13.0 Programming Contest Oct 2019
Contestant *Online*
- Ranked 67th globally and 2nd among teams participating from Turkey.

Turkish Programming Contest (TPC) Sept 2019
Contestant *Istanbul, Turkey*
- Represented Koç University and ranked 12th among teams participating from all over Turkey.

SKILLS

- **Technical**: C++, Python, Pytorch, Deep Learning, Agile Development, Algorithms & Data Structures, Competitive Programming, Data Analysis, Blockchain.
- **Languages:** Arabic (native), English (fluent), French (intermediate), Turkish (intermediate).

PROJECTS

For list of projects you may visit project page in my website.

EXTRACURRICULAR ACTIVITIES

Koç University Programming Championship (KUPC) Jan 2020 - Sept 2020
Problem Setter & Chief Judge *Istanbul, Turkey*
- KUPC is an ACM-ICPC style programming competition for Koç University students.
- Created and tested problems, and prepared competition system.

Koç University ACM Student Chapter (KUACM) May 2019 - Aug 2020
Vice President *Istanbul, Turkey*
- Managed and organized programs and activities carried out under that chapter's supervision.

Koç University Algorithms Program (KUAP) Jan 2019 - Sept 2020
Founder & Coordinator & Instructor *Istanbul, Turkey*
- Created a curriculum that covers basics of problem solving using algorithms and data structures.
- Led discussions and problem solving sessions demonstrating concepts and solutions to problems.

inzva Algorithmic Competition Winter Camp 2019 Jan 2019
Participant & Contestant *Istanbul, Turkey*
- Ranked 1st in final day contest after a week long intensive training on algorithms and data structures.